



CYBERSECURITY IMPLICATIONS OF QUANTUM COMPUTING AND ITS COMBINED USE WITH ARTIFICIAL INTELLIGENCE

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Abstract:

Quantum computing presents new market opportunities, but also significant challenges, particularly in areas such as cybersecurity. Although still in its infancy, quantum computing is maturing rapidly. If quantum computers can provide countries with new offensive capabilities - often in combination with artificial intelligence - it is arguably investment in the quantum communications agenda - in particular quantum networks - that can provide measures and actions to improve cyber defence.

Keywords: Quantum computing, technology policy, cybersecurity, encryption, quantum machine learning.

Título en Español: *Implicaciones para la ciberseguridad de la computación cuántica y su uso combinado con la inteligencia artificial.*

Resumen:

La computación cuántica presenta nuevas oportunidades de mercado, pero también importantes retos, sobre todo en ámbitos como la ciberseguridad. Aunque todavía está en pañales, la computación cuántica está madurando rápidamente. Si los ordenadores cuánticos pueden proporcionar a los países nuevas capacidades ofensivas -a menudo en combinación con la inteligencia artificial-, podría decirse que es la inversión en la agenda de las comunicaciones cuánticas -en particular las redes cuánticas- la que puede proporcionar medidas y acciones para mejorar la ciberdefensa.

Palabras Clave : *Computación cuántica, política tecnológica, ciberseguridad, cifrado, aprendizaje automático cuántico.*

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1. Introduction

The rapid development of quantum technologies offers unprecedented challenges for global security. Quantum technologies exploit the fundamental principles of quantum mechanics², such as the principle of superposition of entanglement, to upgrade and investigate new applications. These “quantum technologies” create new advantages in terms of scope (e.g. sensing), speed (e.g. computation), and processes (e.g. communications) that are much superior to the most advanced “classical” technologies.

Quantum computers will be able to perform operations faster than “classical” supercomputers, quantum networks will allow for the exchange of ultrasecure communications, and quantum sensors will allow militaries to operate with unprecedented accuracy. For these reasons, there is already literature linking quantum technologies to a new revolution in military affairs³.

Since 2019, the year in which Google claimed⁴ “quantum supremacy” (later refuted⁵), attention to quantum technologies has been growing. In the European Union, several countries published national quantum strategies and the competition between the U.S. and China has increased in this realm. The Snowden revelations, export controls^{6 7} and the decoupling of digital supply chains motivated China to develop its native quantum industry, currently offering alternatives to U.S.-made chips and leading the way in some quantum subfields.

This article aims to provide an overview of the situation in quantum technology policy in the United States, China, and the European Union and zoom into quantum cybersecurity policy, including the intersection of quantum and artificial intelligence. The impact of quantum computing on cybersecurity serves often as a thermometer to assess the level of maturity of quantum technologies with claims of new achievements appearing often in press and scientific journals^{8 9}.

Moreover, it shows the value of quantum computing as an offensive capability. Therefore, seen in parallel with developments in the field of quantum communications to defend against quantum attacks, the field of cybersecurity can be a great case study for a reflection on the impact of quantum technologies in international relations, the global security landscape, and, particularly, how quantum technologies can affect the offence-defence balance¹⁰.

² Jaeger, Lars (2018): *The Second Quantum Revolution: From Entanglement to Quantum Computing and Other Super-Technologies*. 1st ed., Cham, Springer International Publishing, at <https://link.springer.com/book/10.1007/978-3-319-98824-5>

³ Der Derian, James; Wendt, Alexander, “Quantizing international relations: The case for quantum approaches to international theory and security practice”, *Security Dialogue* Vol.1, n° 15 (2020)

⁴ Arute, Frank et. al: “Quantum supremacy using a programmable superconducting processor” *Nature*, n° 574 (October 2019), at: <https://www.nature.com/articles/s41586-019-1666-5>

⁵ Kalai, Gil et al.: “Questions and concerns about Google’s Quantum Supremacy Claim”, 3 May 2023, at <https://arxiv.org/pdf/2305.01064>

⁶ “Department of Commerce Releases Export Controls on Quantum Technologies”, U.S. National Quantum Initiative, at <https://www.quantum.gov/department-of-commerce-releases-export-controls-on-quantum-technologies/>

⁷ “Exporting Dual-Use Items”, DG TRADE, European Commission at: https://policy.trade.ec.europa.eu/help-exporters-and-importers/exporting-dual-use-items_en

⁸ For example, at the time of finishing this article, Chinese scientists have claimed to break RSA encryption using D-Wave’s quantum annealer machines. See: Swain, Gyana: “Chinese researchers break RSA encryption with a quantum computing”, 12 October 2024, at: <https://www.csoononline.com/article/3562701/chinese-researchers-break-rsa-encryption-with-a-quantum-computer.html>

⁹ “Google Quantum AI and Collaborators. Quantum error correction below the surface code threshold”, *Nature*, 9 December 2024, at <https://doi.org/10.1038/s41586-024-08449-y>

¹⁰ Jervis, Robert: “Cooperation Under the Security Dilemma.” *World Politics* Vol. 30, n° 2 (1978) pp. 167-214.



This article is divided as follows. Section 2 contextualises the discussion offering a description of sub-fields that make up the field of quantum information sciences: quantum computing and quantum communications. It will describe the main characteristics and will provide concrete examples of applications. Section 3 will create a window through which to look into the quantum agendas of China, the United States, and the European Union offering valuable comparative data such as public investment into quantum technologies or global patent distribution. Lastly, Section 4 offers a review of quantum cybersecurity challenges and their responses, including an overview of risks and opportunities for quantum AI.

2. Brief overview of quantum technologies relevant to cyber defence

Quantum technologies are increasingly recognised as a pivotal area in global technological competition, deeply intertwined with geopolitics. As States strive to harness their transformative potential, there are already foreseen disruptive applications in a variety of sectors, from health and pharma with the discovery of new drugs and chemicals, to financial modelling and artificial intelligence. In fact, the combination of quantum computing and artificial intelligence is poised to boost advancements towards artificial general intelligence¹¹.

2.2 Quantum computing

Quantum computers perform operations on *qubits* or quantum bits that have different characteristics from the classical units of information because of their mechanical properties. Digital data is stored in *bits*, binary digital units that assume either a 1 or 0 state. Quantum bits (*qubits*) differ from classical bits in that they are neither 0 nor 1. They are in a superposition state, roughly 0 and 1 at the same time.

Only when the scientist performs a measurement, or “reads” them one can know the state. Moreover, qubits can be liaised to act as if they were twins even at far distances. This principle, entanglements, works “as if each quantum bit knows what the others are doing and is directly affected by their behaviour”¹². Combined, the natural superposition of states of quantum bits and the entanglement principle allow for the parallelisation of operations¹³, which results in the increase of computational power.

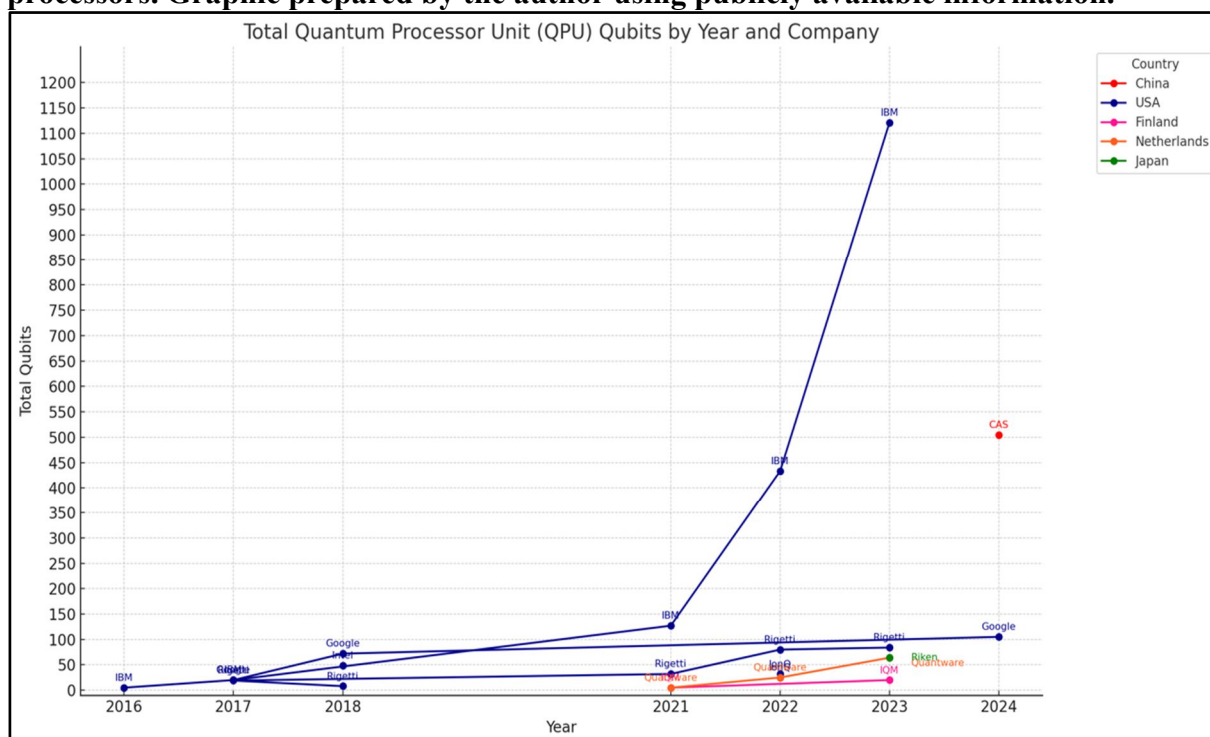
Quantum computing is still in an experimental state. Yet, quantum machines are already in use to improve optimisation problems and in combination with high-performance computers (HPC) to help solve problems efficiently that supercomputers could not solve before or could with a lot of effort.

¹¹ Kosaraju, Deekshitha: "Quantum Computing and Artificial Intelligence: A Fusion Poised to Transform Technology", *International Journal of Research and Review*, Vol. 8, n° 9 (September 2021), pp. 588-593. Available at: https://www.ijrrjournal.com/IJRR_Vol.8_Issue.9_Sep2021/IJRR074.pdf

¹² Jaeger, *op. cit.* p. 44

¹³ *Ibid.*

Figure 1: Advancements in total qubit count of commercially available quantum processors. Graphic prepared by the author using publicly available information.



As technology and industry mature, quantum computers will play an important role in new discoveries like new drugs, chemicals or materials or improving the efficiency of current processes. All these cases open the door to new applications in the field of security and defence: from crafting new sophisticated bioweapons to improving technological advantage in land, sea, and air by the implementation of lighter, more robust, adaptable materials.

2.3 Quantum communications

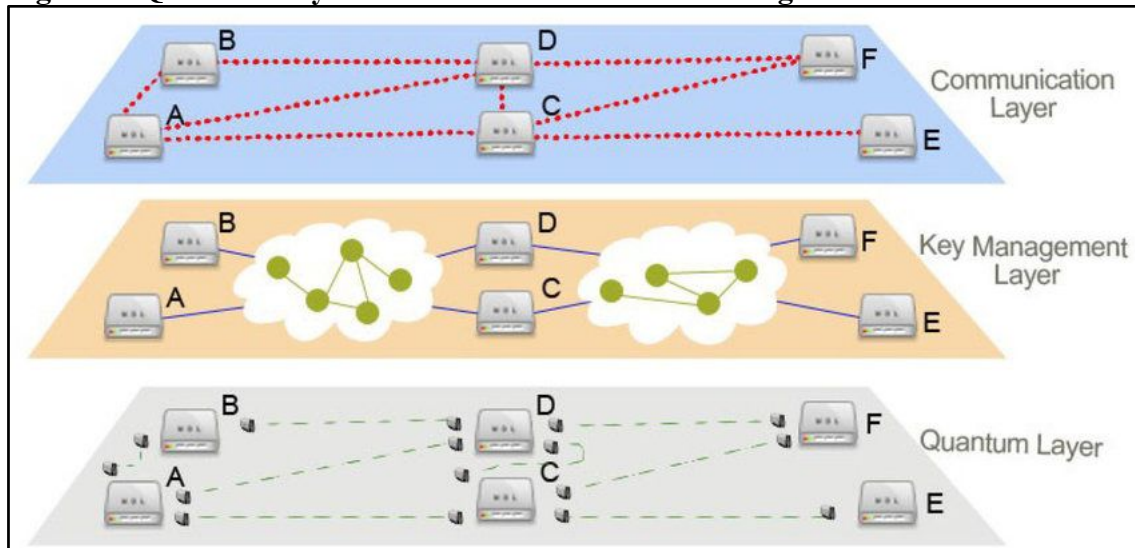
Quantum communications use quantum mechanics to secure communications. Using technologies like quantum key distribution (QKD) and quantum repeaters to solve the distance problems of quantum entanglement, these networks enable information sharing in quantum states between different points of connection (nodes) though there are still questions around their technological maturity¹⁴ (see: Box 1).

A feature of quantum networks is the use of quantum key distribution (QKD) which allows two parties to share encryption keys securely (see: Figure 2). Parties establish a quantum link between the nodes using principles like quantum entanglement, entangled particles in the node generate secure encryption keys (*quantum layer*). These keys are distributed and shared between the nodes (*key management layer*). The keys are used for communication and to allow the encrypted data to be transmitted securely across the network (*communication layer*).

¹⁴ Position Paper on Quantum Key Distribution, at

<https://www.forsvarsmakten.se/contentassets/f7199ed1b90f41529b76970bdb5fcelc/position-paper-on-quantum-key-distribution.pdf>

Figure 2: Quantum key distribution network. Ground segment¹⁵.



Because of the properties of quantum bits, data shared in QKD networks cannot be copied. Unlike in classical networks in which data can be copied and stored away, the uniqueness of qubits prevents information theft. Moreover, because any disturbance can be felt by the network in case of risk of interference communication can be stopped. This offers a new leap in cybersecurity.

Terrestrial QKD segments rely on fibre optic networks. Limitations in entanglement technology and qubits' sensitivity to interference limit the extent of quantum networks (approx. 200km), making necessary the use of repeaters, trusted nodes, and space links.

In the European Union, the quantum communication infrastructure (EUROQCI¹⁶) counts on the participation of the EU 27 member States. Member States are developing QKD networks inside their territory and there are already some examples of cross-border communications using quantum networks¹⁷, but most of the efforts are still within cities and regions. The United States¹⁸ and NATO¹⁹ are only now starting to experiment with quantum networks, leveraging the know-how of their allies and, in the case of the United States, its capacity to mobilise private investments.

Space segments are necessary to overcome the limitations of ground-based links. Space links are necessary to transmit quantum keys for distances beyond 100-200km. Following the example above, in the European Union this will be done in partnership with the European Space Agency (ESA) through the launch of a constellation of quantum satellites (IRIS2). The first satellite will be launched into low earth orbit in 2025²⁰.

¹⁵ Mehic, Miralem; Niemiec, Marcin; Rass, Stefan et al.: "Quantum Key Distribution: A Networking Perspective", *ACM Computing Surveys*, Vol. 53, n° 41 (2020)

¹⁶ "Quantum Communication Infrastructure (EuroQCI)", European Health and Digital Executive Agency, at https://hadea.ec.europa.eu/programmes/connecting-europe-facility/about/quantum-communication-infrastructure-euroqci_en

¹⁷ Stanley, Manoj et al.: "Recent Progress in Quantum Key Distribution Network Deployment and Standards", *Journal of Physics: Conference Series*, n° 2416, at https://www.researchgate.net/figure/Inter-governmental-QKD-networks-between-Italy-Slovenia-and-Croatia-The-network-consists_fig3_366434266

¹⁸ "Quantum Key Distribution Network", DARPA, at <https://www.darpa.mil/about-us/timeline/quantum-key-distribution-network>

¹⁹ NATO, *op. ed*

²⁰ "Eagle-1", European Space Agency, at https://www.esa.int/Applications/Connectivity_and_Secure_Communications/Eagle-1



China remains the most advanced actor in the field of quantum communications. In 2017, China launched into space the first quantum satellite, ‘Micius’, to establish long-distance quantum communications²¹, and in 2023 data showed that ground links exceed 10,000km in length with commercial services available in cities like Beijing or Shanghai²². QKD networks are used nowadays for encrypted calls or security-enhanced financial services.

3. Investment in quantum technologies

Quantum technologies are developing rapidly. Though access to quantum capabilities is currently restricted to a few technologically advanced nations (see: Figure 3), countries have been investing heavily in quantum research and innovation with some publishing dedicated national quantum strategies (see: Table 2). Increasingly, these investments have moved from fostering research excellence to achieving relative strategic gains in the field, reducing strategic vulnerabilities in key aspects of the value chain, and advancing towards technological advantage.

Table 1: Comparison of national quantum strategies²³.

Country	National Strategy	Public funding (€)	Years
China	14 th five-year plan 2021*	~13,8 billion	2021-2025
United States	National Quantum Initiative, 2018	3,4 billion	2019-2023
UK	National Quantum Strategy, 2023	2,91 billion	2023-2033
Germany	Economic Stimulus Package, 2020	2 billion	2021-2025
France	France National Quantum Strategy, 2022	1,05 billion	2021-2025
Netherlands	Quantum Delta NL, 2019	615 million	2021-2028
Denmark	Danish Strategy for Quantum Technology, 2023	150 million	2024-2027
Ireland	Quantum 2030, 2023	budget not disclosed	2023-2030

3.1 China

In China, several policies document the country’s interest in quantum technologies. Already in 2006, the National Medium and Long-Term Science and Technology Development Plan²⁴ emphasised the importance of research in the field of quantum communications. The revelations of former NSA contractor Edward Snowden of a global surveillance network undertaken by the

²¹ Popkin, Gabriel: "China's Quantum Satellite Achieves 'Spooky Action' at Record Distance", *Science*, 15 June 2017, at <https://www.sciencemag.org/news/2017/06/china-s-quantum-satellite-achieves-spooky-action-record-distance>

²² Qi, Wei (2023): "A Brief Introduction to the Latest Progress of China's QKD Industry", ETSI/IQC Quantum Safe Cryptography Event, 14 February 2023, at https://docbox.etsi.org/Workshop/2023/02_QUANTUMSAFECRYPTOGRAPHY/TECHNICALTRACK/WO RLDTOUR/CASQUANTUMNETWORK_QI.pdf

²³ G. Rodríguez, Andrea.: "Finland in the Race for Quantum Advantage: Policy Recommendations for a National Quantum Strategy", The Finnish Innovation Fund, SITRA, May 2024, at <https://www.sitra.fi/wp/wp-content/uploads/2024/05/sitra-finland-in-the-race-for-quantum-advantage.pdf>

²⁴ "Innovation Strategy of China", at https://en.most.gov.cn/pressroom/200706/t20070613_50394.htm

United States further emphasised the idea in China of the need to invest in ultrasecure communications using quantum networks²⁵.

China's 13th Five-Year Plan (FYP), in development at the time of the Snowden revelations, elevated quantum as a major science and explicitly called for the development of quantum communications as a “strategic industry”²⁶. The 14th Five-Year Plan, in place until the end of 2025, further deepens the discussion about the importance of quantum technologies. The 14th FYP calls on specific actions to deepen Chinese leadership in quantum technologies, for example in the fields of standardisation and patents.

Figure 3: Quantum Technology Patent Share (2000-2022).

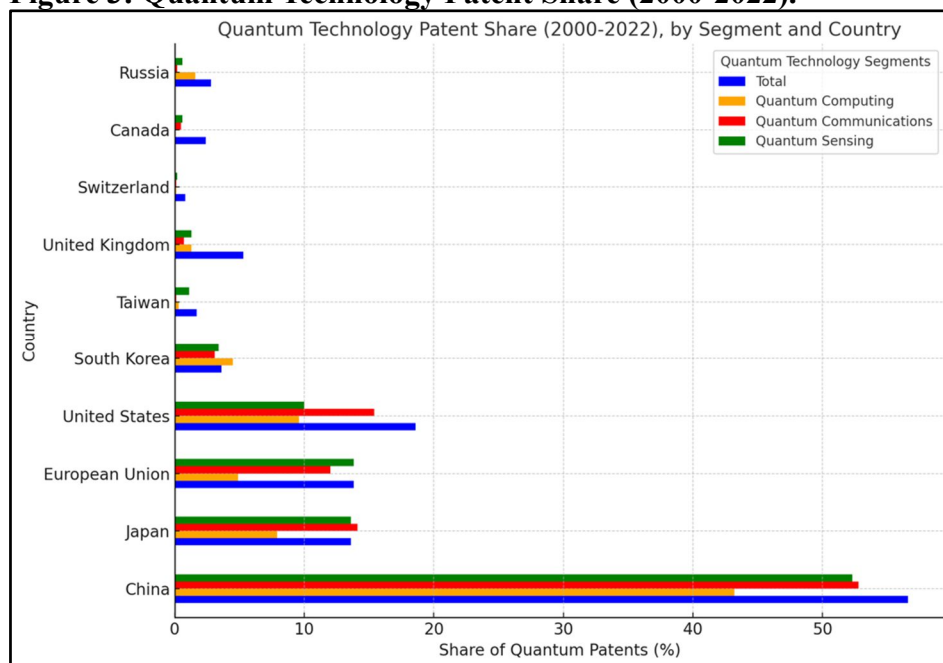


Figure 3 illustrates the global distribution of quantum technology patents until 2022 representing various countries and by technological quantum sub-field. China's dominance is striking with 56,6% of total quantum technology patents with significant leverage across all segments (43,2% in quantum computing, 52,8% in quantum communications and 52,3% in quantum sensing). This advantageous position demonstrates Chinese ambitions and its ability to sustain long-term investments with political direction.

Though a leader in quantum communications, in recent years China has started to pivot towards quantum sensing and quantum computing. Sanctions and trade tensions between the U.S. the EU, and China justify this shift. In April 2024, China unveiled out of the blue a 504-qubit chip, Xiaohong, the second biggest commercially available after IBM's Condor (1,121-qubit) (see: Figure 1)

3.2 United States

In the United States, the 2018 National Cyber Strategy identified for the first time the disruptive impact of quantum technologies on national security and established the need for the U.S. to

²⁵ Qi, Ciel: “China's Quantum Ambitions: A Multi-Decade Focus on Quantum Communications”, *Yale Journal of International Affairs*, 2 February 2024, at <https://www.yalejournal.org/publications/chinas-quantum-ambitions>

²⁶ People's Republic of China: “The 13th Five Year Plan for Economic and Social Development of The People's Republic of China”, March 2016, Beijing, Central Compilation & Translation Press.

implement quantum-proof solutions²⁷. The Strategy compiled and gave structure to previous decentralised efforts of the country to maintain its leverage in new emerging technologies.

Two months after the publication of the Strategy, the U.S. published the National Quantum Initiative Act “to accelerate quantum research and development for the economic and national security of the United States”²⁸. Though the successful launch of Micius in 2017 did have a wakeup call effect in the U.S., the Initiative has become a framework to coordinate federal efforts in the development of quantum technologies as well as to supervise the direction of the such.

Under the umbrella of the initiative, the United States has developed a whole-of-government approach to quantum technologies across three pillars: research and innovation, broadening industrial capabilities, and security and defence²⁹ mobilising more than €3 billion to bridge sovereignty gaps and advance leadership across the three technological areas.

Moreover, the U.S. has deployed a network of 14 centres of excellence across the country to fund quantum programmes and build up the quantum workforce in collaboration with federal departments, but also NASA, the Department of Defence, and the intelligence community. As the United States recognises quantum technologies as an essential element for security and prosperity, many cross-programme initiatives allow for the creation of new use cases also using the experience of defence innovation in the country, such for example the intertwining of the quantum industrial efforts with the 2022 U.S. Chips and Science Act³⁰.

3.3 European Union and member States

In 2016, the EU launched a Quantum Manifesto that recognised the importance of quantum technologies for the future of the European economy. Two years later some of the premises materialised in the Quantum Flagship initiative that counts with independent funding of €1 billion to support research and innovation in quantum technologies until 2028³¹. Beyond the Flagship, the EU funds quantum technologies through several programmes, including QuantERA to which non-EU countries have access to funding like Israel, Turkey, Switzerland of the UK, Accelerator or Horizon Europe.

Moreover, in recent years mentions about quantum technologies in key policy and strategic documents have multiplied. The 2022 Strategic Compass mentioned the importance of quantum-safe communications to ensure resilience. The 2022 Digital Compass established as a target for Europe to have a fully functional quantum computer in 2030. The Chips Act established funding opportunities for quantum chips, and the 2023 Economic Security Strategy established it as one of the four priority areas out of the 10 strategic technologies identified.

However, different perceptions of EU leaders of the strategic importance of quantum technologies and the share of competencies between member States and the EU have impeded the EU from having a bigger role in quantum technologies. While some countries, such as the Netherlands or Germany, see quantum as a national priority it barely shows on the agenda of other smaller countries. These different perceptions risk deepening the asymmetries between EU member States by providing a first-mover advantage to those who accumulate critical resources and services, investment and talent³². As of mid-2024, only five countries have

²⁷ “National Cyber Strategy of the United States of America”, White House, March 2023, at <https://www.whitehouse.gov/wp-content/uploads/2023/03/National-Cybersecurity-Strategy-2023.pdf>

²⁸ *National Quantum Initiative Act*, U.S. Congress, preamble.

²⁹ <https://www.quantum.gov/wp-content/uploads/2022/04/NQI-Factsheet.pdf>

³⁰ “Quantum in the CHIPS and Science Act” U.S. National Quantum Initiative, at <https://www.quantum.gov/quantum-in-the-chips-and-science-act-of-2022/>

³¹ *EU Quantum Flagship*, at <https://qt.eu/about-quantum-flagship/>

³² G. Rodriguez, 2024, *op. cit.*

developed national quantum strategies (the Netherlands, Germany, France, Denmark and Ireland), though more are expected to follow.

4. The Quantum Risks to Cybersecurity

Digitalisation increasingly integrates infrastructure, devices, and services into digital networks. An example of how quantum computing affects international security can be found in its impact on cybersecurity.

Communications in cyberspace are instantaneous, as opposed to non-virtual networks. In cyberspace, geography becomes irrelevant and integrated into one singular digital layer that is ever-growing³³. Because of these two characteristics, States “have more capacity to exercise hard and soft power in cyberspace than in many more traditional domains of world politics³⁴”.

Malicious actors can take advantage of interconnectedness and of the decentralised architecture of cyberspace to perform attacks with a cascading impact on everything on the network. In 2010, a targeted worm attack on Iran’s nuclear plant of Natanz coordinated by Israel and the United States destroyed more than 1,000 centrifuges delaying the Iranian nuclear programme years³⁵. In 2017, the WannaCry ransomware attack affected more than 300.000 terminals across the world, including those of critical services operators such as the UK’s National Health Service³⁶. Months before the US’s National Security Agency found a vulnerability (“Eternal blue”) that was linked to the attack by an APT group. It is arguable that a quantum attack on the network will have similar, if not greater effects.

Table 2: Comparative view of the cyber and physical dimensions

Dimensions	Virtual	Non-Virtual
Temporality	Instantaneous	Not instantaneous
Geography	Integrated and irrelevant	Relevant and accidental
Affordability of attacks	Cheap	Costly
Attribution & Accountability	Hard	Easier
Development	Fluid	Slow

³³ Friedman, Thomas L.: "It's a Flat World, After All", *The New York Times*, 3 April 2005.

³⁴ Nye, Joseph S. (2010): *Cyber Power*, Cambridge, MA, Harvard Kennedy School.

³⁵ Barzashka, Ivanka: "Are Cyber-Weapons Effective?", *The RUSI Journal*, Vol. 158, n° 2 (2013).

³⁶ “NHS England Business Continuity: management toolkit case study: WannaCry attack”, at <https://www.england.nhs.uk/long-read/case-study-wannacry-attack/>



The nature of cyberspace allows attackers to perform cheaper and more powerful attacks than by kinetic means³⁷. This, in addition to the technical hardships in attribution, prosecution, and accountability has made cyberspace a new strategic domain of warfare³⁸.

The effects of quantum computing on cybersecurity tend to be restricted to quantum attacks on encryption. However, advances in technology signal that it is more than the risks to encryption. Quantum computing will have an impact all across the board in cyber operations, including the use of artificial intelligence.

Improvements in the identification, delivery, exploitation, and installation of malware will make cyberattacks more frequent and easier to deploy. Research is still to be done on the capacity of quantum computers to launch multiple petitions at a time, but increased computing speed indicates that this could be the case uncovering new practical cases in the field of botnets and information warfare. Actors without advanced quantum capabilities will be vulnerable to new generations of cyberattacks.

4.1 Quantum attacks on encryption

Quantum computers could speed up the process of producing intelligence by breaking into classical computers to steal information, but they could also retrieve old data. This is already increasing the number of cyber operations in what has been called the “download now, decrypt later” behaviour³⁹. Adversaries can hack into systems now to download encrypted text with the goal of decrypting it once the technology is available. The publication of relevant quantum algorithms in the decade of the 1990s evidence that this will be possible (Shor’s algorithm in 1994 and Grover’s in 1996).

Table 3: Most used cryptographic systems and their resistance to quantum attacks.

Cryptography standard (in-use)	Function	Post-quantum security level	Examples of today’s use
RSA-2048	Encryption & signature	Broken	Internet traffic, including the webpages of all European Institutions, banks, energy, and transport companies.
RSA-3072	Encryption & signature	Broken	VPNs, financial transactions, minimum security level required for intelligence secrets, e-passports.
DH-3072	Key exchange	Broken	Internet protocols such as SSL/TLS, SSH, and IPsec.
256-bit ECDSA	Signature	Broken	Used in Bitcoin and Ethereum exchanges, Companies’ internal communications

³⁷ Nye, *op. cit.*

³⁸ “NATO recognises cyberspace as a ‘Domain of Operations’ at Warsaw Summit”, at <https://ccdcoe.org/incyber-articles/nato-recognises-cyberspace-as-a-domain-of-operations-at-warsaw-summit/>

³⁹ “Quantum-safe Security”, ID Quantique, at <https://www.idquantique.com/quantum-safe-security/overview/>

It is hard to assess what will be the turning point for quantum computers. For example, Fujitsu estimates that 10.000 qubits with 2.23 trillion quantum gates will be necessary to break the RSA-2048 cryptosystems⁴⁰. IBM, the most advanced quantum chipmaker (see: Figure 1) estimates in its roadmap⁴¹ that this will only be possible after 2030. However, a report by the Global Risk Institute estimates that by 2026 1 in 7 cryptosystems will be already impacted by quantum computers and that number goes up to 1 in 2 by 2031⁴².

4.2 Quantum malware and artificial intelligence

In 2023 Generative AI shocked the world with its ability to mimic human speech, though showing limits in reasoning and decision-making⁴³. Advanced AI is already impacting cybersecurity: from pattern recognition to advanced cyber defence to playing a role in designing malware. However, as artificial intelligence and quantum computing develop the convergence of both technologies will have serious implications for global security, including in the cyber domain. This convergence is already happening with the publication of scientific papers and the registration of patents touching on both subfields growing faster than AI or quantum alone⁴⁴. Hence, it is relevant to cover what the effects of the combination of these technologies could mean for the field of cyber defence.

Quantum computing, as a foundational hardware-based technology, affects the software that can be designed and deployed in quantum systems but also gives a comparative advantage to developers designing algorithms that can be run in classical systems—like the ones we use today and will remain the most used even in the quantum era.

The immense computational power of quantum computers compared to classical systems reciprocates with the need for the AI system to access more powerful computing capabilities to train large AI models. For that reason, quantum computing and AI, though different, are also complementary scientific fields.

This complementarity has been a source of research for a long time. Already in 2005, researchers from the Universities of Toronto and Southern California already adventured that quantum computing could power new generations of malware with “no limit to the number and character of possible malware attacks”⁴⁵.

AI-built malware is already challenging traditional cyber defence mechanisms. Contrary to conventional malware that relies on pre-written code and behaves in a fixed manner, AI malware can adapt, learn from the environment it is deployed and optimise real-time attacks. This flexibility challenges traditional malware detection tools, such as antivirus, that are designed to spot unconventional behaviour. Though these tools have increasingly been upgraded with AI, as it happens, it takes the attacker only once to be truly destructive while the defender must be right at all times to guarantee cyber security.

⁴⁰ "Fujitsu quantum simulator assesses vulnerability of RSA cryptosystem to potential quantum computer cryptography threat", Fujitsu, 23 January 2023, at <https://www.fujitsu.com/global/about/resources/news/press-releases/2023/0123-01.html>

⁴¹ "IBM Quantum Roadmap", IBM, at: <https://www.ibm.com/roadmaps/quantum/>

⁴² "Quantum Computing: A New Threat to Cybersecurity", Global Risk Institute at <https://globalriskinstitute.org/publication/quantum-computing-cybersecurity/>

⁴³ "When AI Gets It Wrong: Addressing AI Hallucinations and Bias", MIT Sloan at <https://mitsloanedtech.mit.edu/ai/basics/addressing-ai-hallucinations-and-bias/>

⁴⁴ Coccia, Mario: "Covering Artificial Intelligence and Quantum Technologies: Accelerated Growth Effects in Technological Evolution", *Technologies*, Vol. 12, n° 66, (May 2024).

⁴⁵ Wu, Lian-Ao and Daniel Lidar: "Quantum Malware", *Quantum Information Processing*, Vol. 5, n° 2 (June 2005).

Quantum machine learning⁴⁶ has the potential to affect cybersecurity in several ways. Quantum AI-enhanced malware will increase the ability of already AI-powered malicious code to analyse security protocols and avoid detection by leveraging bigger datasets and responding faster to changes in the environment predicting defensive responses and adapting to them—presumably faster than the AI tools embedded into advanced threat detection protocols.

Moreover, using quantum AI, cyber criminals will be able to find unknown patterns and correlations and use them to target networks more effectively. Precision targeting, which will increase thanks to the ability of quantum computing to process larger datasets than traditional computers, is an area of concern in cybersecurity. Phishing attacks are still the most common to obtain credentials and are increasingly in use to target cloud systems on which multiple companies and even governmental institutions depend at the same time⁴⁷.

While these cases above are only examples of the malicious use of quantum and AI together to increase the lethality of cyberattacks, most research published today is precisely on the use of quantum AI to improve cyber defence protocols. The World Economic Forum already sees quantum machine learning as a “new tool in the cybersecurity locker”⁴⁸, and important sites such as arXiv⁴⁹ or journals such as the multiple ones published by the IEEE⁵⁰ publish increasingly findings in quantum-AI algorithms that can improve threat detection and malware classification. However, the field remains largely experimental and tests in real environments are expensive and offer, under the current state of development, little advantage over traditional AI-powered defence mechanisms and tools.

4.3 Measures Against Quantum Cybersecurity Risks

The parallel processes of significant increases in QPU count (see: Figure 1) and in error correction in quantum computers and developments in quantum networks—though not at the same pace—have paved the way for the development of quantum cybersecurity agendas in different countries leveraging also “classical” solutions, like cryptography. In recent years, the encryption community.

It is important to mention that post-quantum cryptography is not a quantum technology. PQC algorithms are based on mathematical functions that are ready to be implemented in “classical” systems. That is current computers and digital devices. Despite the questions about their robustness once cryptographically significant quantum computers are available, PQC is an urgent and reliable measure to protect digital networks and devices from quantum attacks on encryption in the immediate term.

China’s approach to quantum cybersecurity can be drawn from the country’s investment in quantum networks and the categorisation of quantum communications infrastructure as a national priority. Fuelled by the urgency of the Snowden revelations, as explained in the

⁴⁶ Quantum machine learning (QMA) is the emerging discipline that benefits from quantum advancements to create new artificial intelligence models. QMA builds on quantum data and runs on hybrid quantum-classical mathematical models. “Quantum Machine Learning Concepts”, TensorFlow at <https://www.tensorflow.org/quantum/concepts>

⁴⁷ “CrowdStrike 2024 Threat Hunting Report”, at <https://go.crowdstrike.com/rs/281-OBQ-266/images/CrowdStrike2024ThreatHuntingReport.pdf?version=0>

⁴⁸ Knopf, Christopher and Michael Daniel “Quantum machine learning: a new tool in the cybersecurity locker”, *World Economic Forum*, at <https://www.weforum.org/stories/2023/05/quantum-machine-learning-cybersecurity/> (May 2023).

⁴⁹ E.g. Barrué, Grégoire and Tony Quertier “Quantum Machine Learning for Malware Classification”, June 2023, at <https://arxiv.org/abs/2305.09674>

⁵⁰ E.g. Akter, Shapna et al. “Case Study-Based Approach of Quantum Machine Learning in Cybersecurity: Quantum Support Vector Machine for Malware Classification and Protection”, IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC), Torino, Italy, May-June 2023, pp-1057-1063,

sections above, the country has made significant efforts to create an extensive network of ground and space quantum communications network. Little is known about the Chinese efforts to implement post-quantum encryption though it is highly likely that it would prefer to develop their own native algorithms than following U.S. NIST's recommendations based on national security premises.

At the EU level, it is individual countries the ones in the leadership position with some countries having published detailed roadmaps for the transition to quantum-safe cryptography, detailing migration recommendations and technologies⁵¹. Moreover, in late January 2024, the National Security Agencies of France, Germany, Netherlands and Sweden published a joint white paper in which they called for the transition to post-quantum cryptography noting also the advancements in quantum key distribution⁵². In the course of 2024, this position paper has received letters of support from other EU countries, such as the Czech Republic⁵³.

European Union's action in this field remains scarce and at an early stage in the transition to PQC, but however remains the main actor in funding and coordinating European quantum networks. The publication of the EU's Economic Security Strategy in June 2023 and of the list of strategic technologies in October 2023 was a conversation starter for the strategic importance of quantum networks. Quantum technologies, and under quantum communications, technologies appear as one of the four key priority areas for economic security. Leveraging from advancements in member States, the EU published in April 2024 a Coordinated Implementation Roadmap that mandates member States to create solid strategies.

The United States is arguably ahead in the establishment of a quantum cybersecurity agenda. In 2016, the U.S. National Institute for Standards and Technology (NIST) initiated a process for the standardisation of post-quantum cryptography algorithms. Though there is a need to increase the number, in August 2024 NIST selected three algorithms as "safe" against quantum computers⁵⁴. In addition to this, the U.S. has speeded up the bank of policies aimed at securing network and information systems. In 2022, the U.S. Congress passed the Quantum Cybersecurity Preparedness Act for the migration to post-quantum cryptography. This action complemented a series of White House Memorandums⁵⁵ setting priorities and concrete actions. Since 2023, quantum has been a recurrent topic in the U.S. National Cybersecurity Strategy.

⁵¹ E.g., "Quantum Technologies and Quantum-Safe Cryptography", Federal Office for Information Security, at https://www.bsi.bund.de/EN/Themen/Unternehmen-und-Organisationen/Informationen-und-Empfehlungen/Quantentechnologien-und-Post-Quanten-Kryptografie/quantentechnologien-und-post-quanten-kryptografie_node.html

⁵² Position Paper on Quantum Key Distribution, *op. cit.*

⁵³ "Letter of Support to the Position Paper on Quantum Key Distribution", National Cybersecurity Agency, at <https://nukib.gov.cz/en/infoservis-en/news/2163-letter-of-support-to-the-position-paper-on-quantum-key-distribution/>

⁵⁴ "NIST releases first 3 finalized post-quantum encryption standards", at <https://www.nist.gov/news-events/news/2024/08/nist-releases-first-3-finalized-post-quantum-encryption-standards>

⁵⁵ E.g.: "Memorandum for the Heads of Executive Departments and Agencies: Migrating to Post-Quantum Cryptography" at <https://www.whitehouse.gov/wp-content/uploads/2022/11/M-23-02-M-Memo-on-Migrating-to-Post-Quantum-Cryptography.pdf>



Box 1: Comparison of quantum key distribution and post-quantum cryptography to secure communications⁵⁶.

Quantum Key Distribution vs. Post-Quantum Cryptography

As countries ready their cybersecurity structures for quantum computers, there are still questions about which technologies are better for securing information. To this day, the two most promising are quantum key distribution (QKD) and post-quantum cryptography (PQC) both of them offering a different set of advantages and disadvantages over the other.

Quantum key distribution (QKD) enables two parties to establish a secure communication channel based on quantum physics. Because of the properties of quantum bits (qubits), data shared cannot be copied, which protects against information theft during communications. Moreover, any disturbance or interference in the communication channel could be perceived by the parties that can suddenly decide to stop communicating. This offers a unique advantage against eavesdropping, where a third party “listens” to the conversation.

However, while eavesdropping can be detected, QKD requires pre-sharing encryption keys, which can create an authentication problem. An unauthorised party could potentially supplant the identity of one of the parties (“man-in-the-middle”). Moreover, QKD requires specific infrastructure, which increases the time and cost of the transition, and its sensibility to eavesdropping could increase the risk of denial of service (DoS) cyberattacks. Also, there are still multiple challenges to widespread adoption, such as the distance at which communication can happen (hardly over 200km nowadays) and the need to use trusted nodes to solve this, to go over 200km. For all these reasons, while QKD applications are promising and can add value in the long-term, they are generally perceived as still in the early stages of development.

Post-quantum cryptography (PQC), as a classical computing solution, is a more mature activity area and offers several advantages over quantum key distribution. At the same time, it also has theoretical and practical challenges. PQC can be defined as a set of cryptographic algorithms which are believed to be quantum resistant. These algorithms run on classical hardware, which makes their deployment much faster and cheaper as, in a few words, it would involve little more than a software update. However, PQC protocols have the same vulnerabilities as current cryptographic systems and further technological advancements could allow for the retrospective decryption of these algorithms, hence the reason why the NIST competition is still ongoing. In other words, no practical proof exists that

5. Conclusions

Even though the advent of a cryptographically significant quantum computer might be seen far away, the speed of developments in areas like quantum volume and qubit count in quantum processors show that we are currently in a moment of acceleration. With the advent of quantum computers being used as strategic instruments, countries are rushing to develop national quantum agendas and defensive capabilities like quantum networks to mitigate the effects of quantum computing on cybersecurity.

The development of quantum technologies is closely tied to geopolitical competition. China, the United States and the European Union are competing for funding and talent, and in the race to develop technologically significant capabilities. The capacity to sustain political direction across years gives the advantage to China in quantum communications though the EU is following closely with the EuroQCI network. Still, fragmentation and competing national interests can play against the European Union in leading the way.

The United States has been successful in mobilising policy and investment after 2018 to bridge the gap with China and the EU. However, though leadership is apparent in the field of quantum computing, the announcement of China’s 504 qubit chip is evidence that competition is following closely.

⁵⁶ G. Rodríguez, 2023, *op. cit.*, p. 5.



In the cybersecurity realm, this materialises in two ways. First, the Snowden revelations served as a wake-up call for Chinese leaders to invest in defensive cyber capabilities in light of the future development of quantum computers that could increase the U.S.'s surveillance capacity. This, materialised in the subsequent mentions in the 13th and 14th Five Year Plan has provided China with sufficient advantage. It is arguable that leading quantum communications will only increase uncertainty⁵⁷ of behind China's moves and fuel further competition.

Moreover, promising applications of quantum machine learning might further fragment global collaboration in emerging technologies and hinder data flows, but also the relocation of talent relevant for AI for its implications on quantum and viceversa.

All in all, advancements in standardisation, implementation of quantum solutions, and the creation of industry-relevant use cases, in combination with other technologies like AI, will characterise the new phase of development of quantum technologies, with applications leaving the lab and being deployed in a real environment. Only then these innovations will be able to show their true strategic value.

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⁵⁷ Buchanan, Ben (2016) *The Cybersecurity Dilemma*, London, Hurst.



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